Logistic Regression

Problem Setup

Linear regression deals with finding the expected value of an explained variable y given some explanatory variable x where $y \in \mathbb{R}$. Now consider the case where y is a **boolean variable** in $\{0, 1\}$.

Similar to the regression case, we consider a training set:

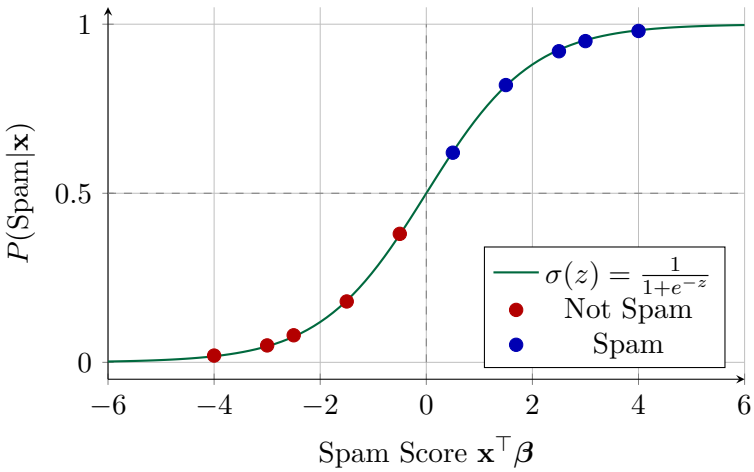
$$D_n := \{(X_i, Y_i)\}_{1 \leq i \leq n}$$

The data points (X_i, Y_i) are i.i.d. random variables following a distribution $\mathcal{P}$ on $\mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} = \mathbb{R}^d$ and $\mathcal{Y} = \{0, 1\}$.

Practical Example: Email Spam Classification

Consider the task of classifying emails as spam ($Y = 1$) or not spam ($Y = 0$). Given features $\mathbf{x}$ extracted from an email (word frequencies, sender information, etc.), we want to predict the probability that the email is spam.

Sigmoid Function: Mapping Scores to Probabilities



The sigmoid function maps any real-valued score to a probability between 0 and 1. Emails with high spam scores are classified as spam; those with low scores are not. The decision boundary occurs at $\mathbf{x}^\top \boldsymbol{\beta} = 0$, where $P(\text{Spam}) = 0.5$.

Motivation for the Logistic Transformation

This setup is similar to linear regression, except that our target function $y = p(x)$ is **not linear**. We need to find a way to view this new function as a linear function, subject to two constraints:

Boundedness. The function $p(x)$ must be bounded between 0 and 1, since it represents a probability.

Diminishing sensitivity. The change in $p(x)$ is not exactly linear in x. If our prediction $p(x)$ is already large (close to 1), changing x by a small amount should not affect the final prediction as much as if $p(x)$ were close to 0.5. In other words, if we are confident in our prediction, we need a larger move in x to change our prediction significantly.

Constructing the Transformation

How can we make a function that is bounded asymptotically in y and requires a multiplicative change in the output to reflect a linear change in the input?

A natural choice is the **logarithm** – but this is only bounded in one direction. To bound it in **both directions**, we use the **logistic (logit) transformation**:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

Hence the term **logistic regression**.

Deriving the Logistic Function

Setting the logit equal to a linear function of x:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta}$$

Solving for p:

$$\frac{p}{1-p} = e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}} \tag{1}$$

$$p = (1-p) \cdot e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}} \tag{2}$$

$$p = e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}} - p \cdot e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}} \tag{3}$$

$$p\left(1 + e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}}\right) = e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}} \tag{4}$$

$$p(\mathbf{x}; \beta_0, \boldsymbol{\beta}) = \frac{e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}}}{1 + e^{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}}} = \frac{1}{1 + e^{-(\beta_0 + \mathbf{x}^\top \boldsymbol{\beta})}} \tag{5}$$

This is the **sigmoid function** (or logistic function), denoted $\sigma(z) = \frac{1}{1+e^{-z}}$.

Remark 1. *The overall specification is much easier to grasp in terms of the transformed probability (the logit) than in terms of the untransformed probability.*

Decision Boundaries and Classification

To minimize the misclassification rate, we predict $Y = 1$ when $p \geq 0.5$ and $Y = 0$ when $p < 0.5$. This means predicting 1 whenever $\beta_0 + \mathbf{x}^\top \boldsymbol{\beta} \geq 0$, and 0 otherwise.

Result: Logistic regression gives us a **linear classifier**.

The Decision Boundary

The **decision boundary** separating the two predicted classes is the solution of:

$$\beta_0 + \mathbf{x}^\top \boldsymbol{\beta} = 0$$

This is a **hyperplane** in $\mathbb{R}^d$. Finding the logistic regression parameters is therefore **equivalent to finding a linear classifier**.

Distance from the Decision Boundary

We derive the signed distance from a point $\mathbf{x}$ to the decision boundary.

Setup: The decision boundary is $\{\mathbf{z} : \boldsymbol{\beta}^\top \mathbf{z} + \beta_0 = 0\}$. The unit normal vector to this hyperplane is $\frac{\boldsymbol{\beta}}{|\boldsymbol{\beta}|}$.

Derivation: Let $\mathbf{x}_0$ be any point on the decision boundary (so $\boldsymbol{\beta}^\top \mathbf{x}_0 + \beta_0 = 0$). The signed distance from $\mathbf{x}$ to the boundary is the projection of $(\mathbf{x} - \mathbf{x}_0)$ onto the unit normal:

$$\text{distance} = (\mathbf{x} - \mathbf{x}_0) \cdot \frac{\boldsymbol{\beta}}{|\boldsymbol{\beta}|} = \frac{\mathbf{x}^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \boldsymbol{\beta}}{|\boldsymbol{\beta}|}$$

Since $\mathbf{x}_0$ is on the boundary, we have $\mathbf{x}_0^\top \boldsymbol{\beta} = -\beta_0$. Substituting:

$$\boxed{\text{distance} = \frac{\beta_0 + \mathbf{x}^\top \boldsymbol{\beta}}{|\boldsymbol{\beta}|}}$$

Interpretation: The class probabilities depend on the **distance from the boundary**. Probabilities approach 0 or 1 more rapidly when $|\boldsymbol{\beta}|$ is larger.

Choice of Loss Function

To minimize the empirical risk, we need to choose a loss function to assess the performance of a prediction.

Binary Loss (0-1 Loss)

A natural loss is the **binary loss**: 1 if there is a mistake ($f(X_i) \neq Y_i$), and 0 otherwise. The empirical risk is:

$$\hat{\mathcal{R}}_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{Y_i \neq \mathbf{1}_{X_i^\top \boldsymbol{\beta} \geq 0}}$$

Problem: This loss function is **neither convex nor differentiable** in $\boldsymbol{\beta}$. The minimization problem $\min_{\boldsymbol{\beta}} \hat{\mathcal{R}}_n(\boldsymbol{\beta})$ is extremely hard to solve (NP-hard in general).

The Logistic Loss

The idea of logistic regression is to replace the binary loss with a **convex surrogate loss**. The **logistic loss** $\ell : \mathbb{R} \times \{0, 1\} \to \mathbb{R}_+$ assigns to a linear prediction $\hat{y} = \mathbf{x}^\top \boldsymbol{\beta}$ and an observation $y \in \{0, 1\}$ the loss:

$$\ell(\hat{y}, y) := y \log(1 + e^{-\hat{y}}) + (1 - y) \log(1 + e^{\hat{y}})$$

Definition 2 (Logistic Regression Estimator)**.** *The **logistic regression estimator** is the solution of:*

$$\hat{\boldsymbol{\beta}}_{logit} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^{n} \ell(\mathbf{X}_i^\top \boldsymbol{\beta}, Y_i)$$

where ℓ is the logistic loss defined in Equation (1).

Advantage over hinge loss: The logistic loss has a **probabilistic interpretation** by modeling $\mathbb{P}(Y = 1|X)$.

Maximum Likelihood Derivation

We now show that **minimizing the logistic loss is equivalent to maximizing the likelihood** under a Bernoulli model.

Probabilistic Model

We model the conditional probabilities using the sigmoid function:

$$P(Y = 1|X = \mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \mathbf{x}) \tag{6}$$

$$P(Y = 0|X = \mathbf{x}) = 1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}) \tag{7}$$

These can be combined into a single expression using the **Bernoulli distribution**:

$$P(Y = y|X = \mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \mathbf{x})^y \cdot [1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x})]^{(1-y)}$$

Log-Likelihood

For i.i.d. data, the **likelihood** is the product over all data points:

$$L(\boldsymbol{\theta}) = \prod_{i=1}^{n} \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)})^{y^{(i)}} \cdot [1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)})]^{(1-y^{(i)})}$$

Taking the **logarithm** (which converts the product to a sum):

$$LL(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left[y^{(i)} \log \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}) + (1 - y^{(i)}) \log(1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)})) \right]$$

Equivalence to Logistic Loss

The **negative log-likelihood** is:

$$-LL(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left[-y^{(i)} \log \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}) - (1 - y^{(i)}) \log(1 - \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)})) \right]$$

Using the identity $\sigma(-z) = 1 - \sigma(z)$, we can write $-\log \sigma(z) = \log(1 + e^{-z})$ and $-\log(1 - \sigma(z)) = -\log \sigma(-z) = \log(1 + e^{z})$.

Substituting with $\hat{y} = \boldsymbol{\theta}^\top \mathbf{x}$:

$$-LL(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left[y^{(i)} \log(1 + e^{-\hat{y}^{(i)}}) + (1 - y^{(i)}) \log(1 + e^{\hat{y}^{(i)}}) \right]$$

This is exactly the **logistic loss** from Equation (1):

$$\boxed{-LL(\boldsymbol{\theta}) = \sum_{i=1}^{n} \ell(\hat{y}^{(i)}, y^{(i)})}$$

Conclusion: Minimizing the empirical risk with logistic loss $\Leftrightarrow$ Maximizing the log-likelihood under a Bernoulli model.

Gradient Computation

Derivative of the Sigmoid Function

First, we compute the derivative of $\sigma(z)$ with respect to z:

$$\frac{\partial}{\partial z}\sigma(z) = \frac{\partial}{\partial z}\left(\frac{1}{1+e^{-z}}\right)$$

Using the quotient rule:

$$= \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z) \cdot (1-\sigma(z))$$

$$\boxed{\frac{\partial}{\partial z}\sigma(z) = \sigma(z)(1-\sigma(z))}$$

Derivative with Respect to Parameters

Using the chain rule:

$$\frac{\partial}{\partial \theta_j}\sigma(\boldsymbol{\theta}^\top \mathbf{x}) = \frac{\partial \sigma}{\partial z} \cdot \frac{\partial z}{\partial \theta_j}$$

where $z = \boldsymbol{\theta}^\top \mathbf{x}$.

Since $\frac{\partial z}{\partial \theta_j} = x_j$:

$$\boxed{\frac{\partial}{\partial \theta_j}\sigma(\boldsymbol{\theta}^\top \mathbf{x}) = \sigma(\boldsymbol{\theta}^\top \mathbf{x})(1-\sigma(\boldsymbol{\theta}^\top \mathbf{x})) \cdot x_j}$$

Gradient of the Log-Likelihood

Starting with the log-likelihood for a single data point:

$$\frac{\partial}{\partial \theta_j}\left[y \log \sigma(\boldsymbol{\theta}^\top \mathbf{x}) + (1-y)\log(1-\sigma(\boldsymbol{\theta}^\top \mathbf{x}))\right]$$

Applying the chain rule to each term:

$$= \frac{y}{\sigma(\boldsymbol{\theta}^\top \mathbf{x})} \cdot \frac{\partial \sigma}{\partial \theta_j} - \frac{1-y}{1-\sigma(\boldsymbol{\theta}^\top \mathbf{x})} \cdot \frac{\partial \sigma}{\partial \theta_j} \tag{8}$$

$$= \left[\frac{y}{\sigma(\boldsymbol{\theta}^\top \mathbf{x})} - \frac{1-y}{1-\sigma(\boldsymbol{\theta}^\top \mathbf{x})}\right] \cdot \frac{\partial \sigma}{\partial \theta_j} \tag{9}$$

Finding a common denominator:

$$= \left[\frac{y(1-\sigma(\boldsymbol{\theta}^\top \mathbf{x})) - (1-y)\sigma(\boldsymbol{\theta}^\top \mathbf{x})}{\sigma(\boldsymbol{\theta}^\top \mathbf{x})(1-\sigma(\boldsymbol{\theta}^\top \mathbf{x}))}\right] \cdot \sigma(\boldsymbol{\theta}^\top \mathbf{x})(1-\sigma(\boldsymbol{\theta}^\top \mathbf{x})) \cdot x_j$$

Simplifying the numerator:

$$y - y\sigma(\boldsymbol{\theta}^\top \mathbf{x}) - \sigma(\boldsymbol{\theta}^\top \mathbf{x}) + y\sigma(\boldsymbol{\theta}^\top \mathbf{x}) = y - \sigma(\boldsymbol{\theta}^\top \mathbf{x})$$

The $\sigma(\boldsymbol{\theta}^\top \mathbf{x})(1-\sigma(\boldsymbol{\theta}^\top \mathbf{x}))$ terms cancel:

$$\boxed{\frac{\partial LL}{\partial \theta_j} = \left[y - \sigma(\boldsymbol{\theta}^\top \mathbf{x})\right] x_j}$$

Full Gradient (All Data Points)

For the entire dataset:

$$\boxed{\frac{\partial LL(\boldsymbol{\theta})}{\partial \theta_j} = \sum_{i=1}^{n} \left[y^{(i)} - \sigma(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}) \right] x_j^{(i)}}$$

In vector/matrix notation:

$$\nabla LL(\boldsymbol{\theta}) = \mathbf{X}^\top (\mathbf{y} - \sigma(\mathbf{X}\boldsymbol{\theta}))$$

where $\mathbf{X} := (\mathbf{X}_1, \ldots, \mathbf{X}_n)^\top$, $\mathbf{y} := (Y_1, \ldots, Y_n)^\top$, and $\sigma(\mathbf{X}\boldsymbol{\beta})_i := \sigma(\mathbf{X}_i^\top \boldsymbol{\beta})$.

Optimization

No Closed-Form Solution

Bad news: The equation $\nabla \hat{\mathcal{R}}_n(\boldsymbol{\beta}) = 0$ has **no closed-form solution**.

It must be solved through **iterative algorithms**: gradient descent/ascent, Newton's method, or stochastic gradient descent.

Convexity Guarantees Unique Solution

Fortunately, optimization is tractable because the logistic loss is **convex** in its first argument.

The second derivative of the loss is:

$$\frac{\partial^2 \ell(\hat{y}, y)}{\partial \hat{y}^2} = \sigma(\hat{y}) \cdot \sigma(-\hat{y}) = \sigma(\hat{y})(1 - \sigma(\hat{y})) > 0$$

Since the second derivative is strictly positive, the loss is **strictly convex**, and therefore the solution is **unique**.

Gradient Ascent Update Rule

To maximize the log-likelihood, we use **gradient ascent**:

$$\boldsymbol{\theta} := \boldsymbol{\theta} + \alpha \nabla LL(\boldsymbol{\theta})$$

where $\alpha > 0$ is the learning rate.

Regularization

Similarly to linear regression, logistic regression may **overfit** the data, especially when $p > n$ (more features than samples).

To prevent overfitting, we add a **regularization term** such as $\lambda |\boldsymbol{\beta}|_2^2$ to the logistic loss, yielding **L2-regularized logistic regression**.

Definition 3 (Regularized Logistic Regression)**.** *The **L2-regularized logistic regression estimator** is the solution of:*

$$\hat{\boldsymbol{\beta}}_{ridge} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^d} \left\{ \frac{1}{n} \sum_{i=1}^{n} \ell(\mathbf{X}_i^\top \boldsymbol{\beta}, Y_i) + \lambda \|\boldsymbol{\beta}\|_2^2 \right\}$$

The regularization parameter $\lambda > 0$ controls the tradeoff between fitting the data and keeping coefficients small. Larger λ produces more shrinkage.

Summary: The Training Algorithm

Putting it all together, the complete logistic regression training procedure is:

Algorithm: Logistic Regression via Gradient Ascent

Input: Training set $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$, learning rate $\eta > 0$
Output: Parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$

```
Initialize θ_j ← 0 for all j = 0, 1, ..., d
repeat
    gradient_j ← 0 for all j = 0, 1, ..., d
    for each training example (x, y) do
        for each parameter j do
            gradient_j ← gradient_j + x_j (y − 1 / (1 + e^{−θ^⊤ x}))
        end for
    end for
    θ_j ← θ_j + η · gradient_j for all j = 0, 1, ..., d
until convergence
return θ
```

The algorithm iterates until the parameters converge (i.e., the change in $\boldsymbol{\theta}$ between iterations falls below some threshold ε) or a maximum number of iterations is reached. The learning rate η controls the step size: too large leads to oscillation, too small leads to slow convergence.